# Riding the Wave?

## Who Starts, Follows and Gains from Machine Learning Research Trends

A Study of 54,579 ICLR Submissions, 2017–2026

Minghan Li

Supervisor: A/Prof Jianzhong Qi

# Outline

1. Motivation and Research Questions
2. Data and Measurement
3. Hindsight Bias in Topic Maps
4. Empirical Strategy
5. Results
6. Held-out Replication
7. Conclusion

# Motivation: Research Fashions in Machine Learning

| Share of ICLR submissions that ... | 2017 | 2026 |
|---|---|---|
| propose a new architecture | 24% | 8% |
| build on a foundation model | 0.3% | 54% |
| carry a branded title ("NAME: subtitle") | 9% | 41% |

Are these fashions? What do they displace, who drives them, and are followers rewarded?

# Research Questions and Hypotheses

| Research question | Key prediction |
|---|---|
| **RQ1 Rise:** what does a fashion displace? | More foundation-model (FM) papers in a topic $\rightarrow$ fewer new-architecture papers (H1d) |
| **RQ2 Following:** who introduces new work? | Newcomers imitate: fewer new-method papers, more benchmark papers (H2b, H2c) |
| **RQ3 Reward:** are followers rewarded? | Crowded topics $\rightarrow$ lower acceptance (H3a) |

All predictions were written down before any test was run (10 hypotheses in total).

# Related Work and Positioning

**Who drives new work? Two views**

- *Fashion and herding:* newcomers imitate (Abrahamson, 1991; Bikhchandani et al., 1992)
- *Outsider innovation:* newcomers bring new ideas (Foster et al., 2015; Azoulay et al., 2019)

**What we add to meta-research on machine learning**

(Koch et al., 2021; Birhane et al., 2022)

- Content trends across all submissions, not only accepted papers
- Topics defined without hindsight
- Tests registered in advance (Nosek et al., 2018), plus a held-out replication

# Data and Measurement

| Construct | Measure |
|---|---|
| Corpus | Provenance-preserving ICLR corpus from our dated official OpenReview snapshots. Existing datasets informed coverage and source audits but were not pooled (Re$^2$: Zhang et al., 2025; Louis archive: MlouisBE, n.d.; Paper Copilot: Yang et al., 2025) |
| Papers | 43,480 submissions, 2017–2026 (accepted, rejected and withdrawn); 11,099 held-out papers for the replication |
| Topic | SPECTER2 embeddings of title + abstract (Singh et al., 2023), clustered into $k = 100$ topics; results agree at $k = 75$ and $150$ |
| Contribution type | LLM with a fixed codebook (method, benchmark, empirical, theory, other); blind second pass $\kappa = 0.93$ |
| Newcomer paper | No author has an earlier ICLR submission in the topic (73% of papers) |
| Acceptance | Final decision; locked away until the single registered run |

# Hindsight Bias in Topic Maps

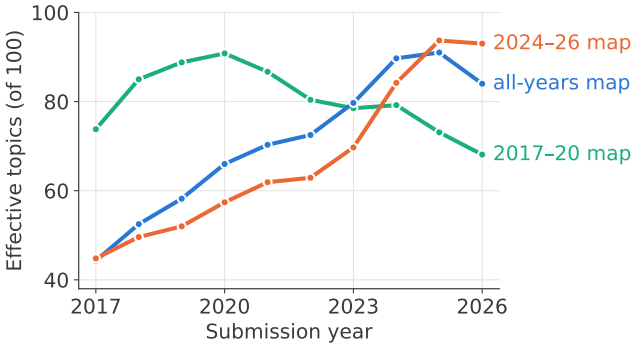



Figure 1: Topic diversity per year under three topic maps.

**Topic map**

The 100 topic clusters, fitted on papers from chosen years; every paper is then assigned to its nearest topic.

Each map finds peak diversity in the years it was fitted on.

⇒ The fitting window shapes the history one sees.

# Consequence: Conclusions Depend on the Map

- Emerging topics ($\geq 3\times$ growth): **8** on the 2017–20 map vs. **29** on the all-years map.
- Do topics age, from new methods towards benchmarks? All-years map: no. 2017–20 map: yes.

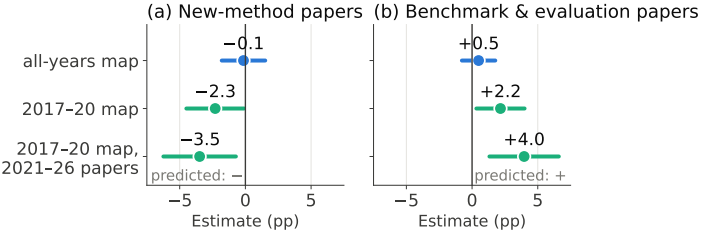



Figure 2: Same papers, same model: coefficient on topic maturity, 95% CI (2017–20 rows exploratory).

$\Rightarrow$ Every registered result must also hold on the 2017–20 map and a year-balanced map.

# Empirical Strategy

**Model** (linear probability)

$$y_i \;=\; \beta\, x_i \;+\; \mu_{\text{topic}(i)} \;+\; \tau_{\text{year}(i)} \;+\; \varepsilon_i$$

- $y_i$: paper type or acceptance (0/1); $x_i$: e.g. newcomer paper
- Papers compared within the same topic and year; standard errors clustered by topic

## Supported only if

1. predicted sign, Holm-adjusted $p < 0.05$
2. same sign in $\geq 75\%$ of alternative specifications
3. same sign on both hindsight checks

## Protocol

Registered in advance, acceptance locked, code frozen, run once; held-out tests registered before their run.

# RQ1: Where Foundation Models Spread, New Architectures Fade

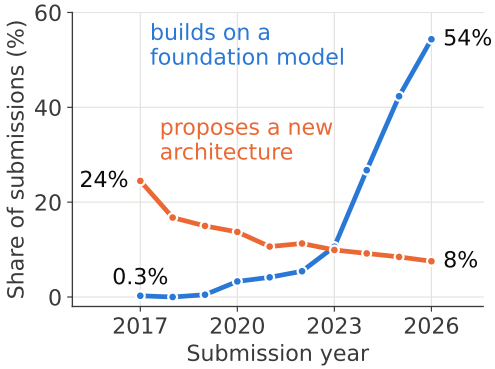



Figure 3: Field-wide trends (descriptive).

Within topic and year, FM adoption 0 → 100%:

| New-architecture papers | Discovery | Held-out |
|---|---|---|
| Change (pp) | $-8.0$ | $-11.0$ |
| 95% CI | $[-12.1, -3.8]$ | $[-18.1, -3.9]$ |

**Supported and replicated**

Topics at 50% vs. 10% FM adoption: about one third fewer new-architecture papers.

# RQ2: Newcomers Bring New Methods, Opposite to Our Prediction

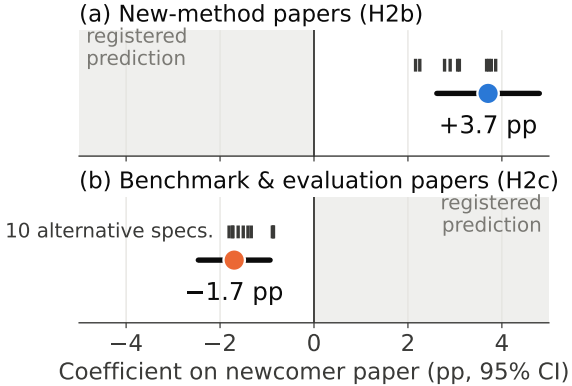



Figure 4: Discovery; within topic and year; grey: predicted side.

| Newcomer paper (pp) | Method | Benchmark |
|---|---|---|
| Predicted | − | + |
| Discovery | +3.7 | −1.7 |
| Held-out | +3.6 | −2.1 |

Reversed predictions registered, then **replicated** on held-out papers.

**Reading**

Consistent with outsider innovation, not with herding.

# RQ3: Is Following the Fashion Rewarded?

| Association with acceptance (pp, 95% CI) | Discovery | Held-out |
|---|---|---|
| Crowded topic, per log unit of prior papers (H3a) | $-1.7$ $[-3.7, +0.3]$ | $-1.5$ $[-5.2, +2.2]$ |
| Branded title (exploratory) | no premium | – |
| New architecture $\times$ topic FM adoption (exploratory) | $+8.3$ $[+1.3, +15.4]$ | – |

Table 1: Within topic and year. Crowding: neither split could detect effects below about 4–5 pp.

## Key point

No detectable reward for following the fashion; papers that go against the tide may even do better (exploratory).

# Do the Results Hold on Held-out Papers?

**Three of four key results replicated; crowding stays inconclusive.**

| Hypothesis | Pred. | Discovery | Held-out [95% CI] | Verdict |
|---|---|---|---|---|
| **H1d** FM adoption → new-architecture papers | − | −8.0 | −11.0 [−18.1, −3.9] | supported |
| **H2b′** newcomer → new-method paper | + | +3.7 | +3.6 [+1.5, +5.7] | supported |
| **H2c′** newcomer → benchmark paper | − | −1.7 | −2.1 [−3.3, −0.8] | supported |
| **H3a** crowded topic → acceptance | − | −1.7 | −1.5 [−5.2, +2.2] | inconclusive† |

Table 2: 11,099 papers from a released former test split, never analysed before one registered run: an out-of-sample replication, not a certified untouched holdout. Supported: one-sided Holm $p < 0.05$; pp. † Too weak to tell (below $\text{MDE}_{80}$).

Also registered: post hoc tests from discovery, e.g. far jumps from a team's own past work go with lower acceptance (replicated). Full table in the report.

# Contributions

1. **Method:** topic maps fitted with hindsight can manufacture trends; an early-year map is a cheap check.
2. **Design:** registered, outcome-blind, single-run protocol, with a held-out replication.
3. **Findings** (associations):
   - where foundation models spread, new architectures fade (replicated)
   - newcomers lean towards new methods, not imitation (replicated)
   - no detectable acceptance reward for following the fashion

# Limitations and Future Work

**Limitations**

- One venue; associations, not causes
- LLM labels: consistent ($\kappa = 0.93$), not yet human-validated
- Held-out papers: a released former split, not certified untouched
- 2021 author-ID change undercounts incumbents

**Future work**

- A prospective test on future ICLR submissions
- ACL Anthology / ARR
- Human validation of contribution labels

# Thank you

Questions and comments welcome

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